

6(a)

structure in the heart	letter
valves preventing back flow of blood into the ventricle	S ;
blood vessel carrying blood under highest pressure	P ;
chamber that pumps blood to the lungs	N ;

3

Question	Answer	Marks
6(b)	<i>treat as neutral effects of nicotine on the brain and ref. to adrenaline</i> <i>any three from:</i> increased heart rate ; I increased heart beat increased blood pressure ; increases risk of, blood clotting / thrombus formation ; A increases risk of thrombosis A increases stickiness of platelets damages the endothelium ; AVP ; e.g. stimulates vasoconstriction / reduces the diameter of blood vessels A named blood vessel types	3
6(c)(i)	<i>any one from:</i> two nuclei vs one ; <i>ref. to</i>, DNA / chromosome, content ; cytokinesis has not occurred ; A cytoplasm has not divided (into two cells)	1
6(c)(ii)	<i>any two from:</i> <i>in context of cardiac myocyte or overall heart function</i> unable to replace, damaged / worn out / old, cardiac myocytes ; R repair myocytes unable to repair (damaged) cardiac, muscle / tissue ; A heart tissue repair (to cardiac muscle may be) with, unspecialised cells / scar tissue ;	2